Ender 3 S1 firmware factory

Personal project · 2026 · Python, JavaScript, embedded C configuration

A build service for custom Marlin firmware

github.com/darthinvader/Ender3S1Firmware · darthinvader.github.io/Ender3S1Firmware

Any upstream checkpoint, any feature selection: compiled, verified, written to the card - with every feature's cost measured in bytes rather than estimated.

28 upstream checkpoints (tags + branches), fetched on demand	**71 / 66** catalogue features / with a measured byte cost	**0 B** size-predictor error on the reference build	**3 / 3** CI workflows green (verify, rebuild, configurator)

WHAT I BUILT

- **The service** - a Python standard-library HTTP server and a vanilla JavaScript interface (no framework, no build step) that takes any upstream Marlin checkpoint, applies a 71-feature selection, queues the compile, streams the log, verifies the ARM vector table and returns the image with its manifest.
- **The measurement harness** - every feature compiled on and off against two baselines to give a real flash/RAM delta; 66 of 71 features measured, which lets the service predict a size to the byte and refuse selections that cannot fit (reference build: 232,344 of 233,472 bytes, prediction exact).
- **The hardware end** - a FAT32 writer that formats the card, renames the image to an unused 8.3 name and md5-verifies every copy, for the mainboard and the display panel, plus the OS/asset pairing rules that keep the panel from being bricked.
- **Quality and automation** - pinned upstream commits, committed binaries with manifests, **make verify** (checksums, vector tables, configuration assertions, link checking, 14 live service checks) and three GitHub Actions workflows, all passing.
- **Recovery engineering** - a display panel bricked during the work; I isolated the mismatched asset pack, documented the recovery and shipped the kit and the evidence trail.

STACK

Python (standard library only) · JavaScript (no framework) · Marlin / embedded C configuration · PlatformIO · STM32F103 · Linux · FAT32 / SD card · GitHub Actions

Upstream: Marlin and mriscoc Professional Firmware, plus Creality display images. Mine: the configuration, measurement harness, build service, SD-card writer, documentation and verification.